LIMITI CON PARAMETRI

$$\lim_{x\to 0} \frac{(\sqrt[4]{1+x^2}-1)\sin^2 x}{(e^{3x^2}-1)^\alpha} \qquad \alpha \in \mathbb{R}$$

$$\lim_{x\to 0} \frac{\frac{1}{4}x^2 \cdot (x)^2}{(3x^2)^\alpha} = \lim_{x\to 0} \frac{\frac{1}{4}x^4}{3^\alpha x^{2\alpha}} =$$

$$(\sqrt[4]{1+x^2}-1) = \left((1+x^2)^{1/4}-1\right) \sim \frac{1}{4}(x^2)$$

$$(1+x)^\alpha - 1 \sim \alpha x$$

$$= \lim_{x\to 0} \frac{1}{4\cdot 3^\alpha} \cdot x^{4-2\alpha}$$

se $4-2\alpha > 0$ $\quad \alpha < 2 \Rightarrow 0$

se $4-2\alpha = 0$ $\quad \alpha = 2 \Rightarrow \frac{1}{4\cdot 3^2} = \frac{1}{36}$

se $4-2\alpha < 0$ $\quad \alpha > 2 \Rightarrow \to +\infty$

$4-2\alpha = 2(2-\alpha)$

$$\lim_{x\to 0} \frac{1-\cos\left(e^{\alpha x}-1\right)}{1-\cos x} \qquad \alpha \in \mathbb{R}$$

$e^{\alpha x}-1 \to 0$

$$1-\cos x \sim \frac{1}{2}x^2 \qquad x \to 0$$

$$\lim_{x\to 0} \frac{\frac{1}{2}\left(e^{\alpha x}-1\right)^2}{\frac{1}{2}x^2} = \lim_{x\to 0} \frac{\frac{1}{2}(\alpha x)^2}{\frac{1}{2}x^2} = \alpha^2$$

$$\lim_{x\to 0^+} \frac{\log\left(x^4+2x^4\right)-\log\left(3x^4+x^2\right)}{\cos\left(\frac{\pi}{2}\cos x\right)}$$

$$\lim_{x\to 0^+} \frac{\log\left(\frac{x^4+2x^4}{3x^4+x^2}\right)}{\cos\left(\frac{\pi}{2}\cos x + \frac{\pi}{2} - \frac{\pi}{2}\right)}$$

$$\lim_{x\to 0^+} \frac{\log\left(\frac{x^4+2x^4}{3x^4+x^2}\right)}{\cos\left(-\frac{\pi}{2}(1-\cos x)+\frac{\pi}{2}\right)}$$

$$\lim_{x\to 0^+} \frac{\log\left(\frac{x^\alpha+2x^4}{3x^4+x^2}\right)}{-\sin\left[-\frac{\pi}{2}(1-\cos x)\right]}$$

$$\lim_{x\to 0^+} \frac{\log\left(\frac{x^\alpha+2x^4}{3x^4+x^2}\right)}{\sin\left[\frac{\pi}{2}(1-\cos x)\right]}$$

$$\lim_{x\to 0^+} \frac{\log\left(\frac{x^\alpha+2x^4}{3x^4+x^2}\right)}{\sin\left(\frac{\pi}{2}\cdot\frac{1}{2}x^2\right)} \qquad \sin x \sim x \quad x\to 0$$

$$\lim_{x\to 0^+} \frac{\log\left(\frac{x^\alpha+2x^4}{3x^4+x^2}\right)}{\frac{\pi}{4}x^2}$$

$\alpha > 4$ $\quad \lim_{x\to 0^+} \log\left(\frac{2x^4}{x^2}\right)\left(\frac{4}{\pi x^2}\right)$

$= +\infty$

$\alpha = 4$ $\quad \lim_{x\to 0^+} \log\left(\frac{3x^4}{x^2}\right)\left(\frac{4}{\pi x^2}\right) = -\infty$

$\alpha < 4 \quad \lim_{x \to 0^+} \log\left(\frac{x^\alpha}{x^2}\right)\left(\frac{4}{\pi x^2}\right)$

$\lim_{x \to 0^+} \log x^{(\alpha-2)} \left(\frac{4}{\pi x^2}\right)$

$\lim_{x \to 0^+} (\alpha-2)\log x \left(\frac{4}{\pi x^2}\right)$

$\alpha = 2 \quad \lim_{x \to 0^+} \log\left(\frac{x^2+2x^4}{3x^4+x^2}\right) \cdot \left(\frac{4}{\pi x^2}\right)$

$\lim_{x \to 0^+} \left(\frac{x^2+2x^4}{3x^4+x^2} - 1\right)\left(\frac{4}{\pi x^2}\right)$

$\lim_{x \to 0^+} \left(\frac{x^2+2x^4-3x^4-x^2}{3x^4+x^2}\right)\left(\frac{4}{\pi x^2}\right)$

$\lim_{x \to 0^+} \left(\frac{-x^4-2}{3x^4+x^2}\right)\left(\frac{4}{\pi x^2}\right) = \frac{4}{\pi}$

$\log x \sim x-1 \quad se \quad x \to 1$

$\log(1+x) \sim x \quad se \quad x \to 0$

$\lim_{x \to 0^+} \left((\alpha-2)\log x\right) \cdot \left(\frac{4}{\pi x^2}\right)$

$2 < \alpha < 4 \quad \to -\infty$

$\alpha < 2 \quad \to +\infty$

CONTINUITÀ

Determinare i valori di a, b per i quali

$$f(x)=\begin{cases}\dfrac{a}{2-x} & x<1\\ b & x=1\\ 2x^2+x+b & x>1\end{cases}$$

è continua nel suo dominio. Rappresentarla nel piano cartesiano. $f(x)$ è definita in $\mathbb{R}$

Studio continuità in $x=1$

$$\lim_{x\to 1^-}\frac{a}{2-x}=a \qquad \lim_{x\to 1^+}2x^2+x+b=2+b+1$$

$$f(1)=b$$

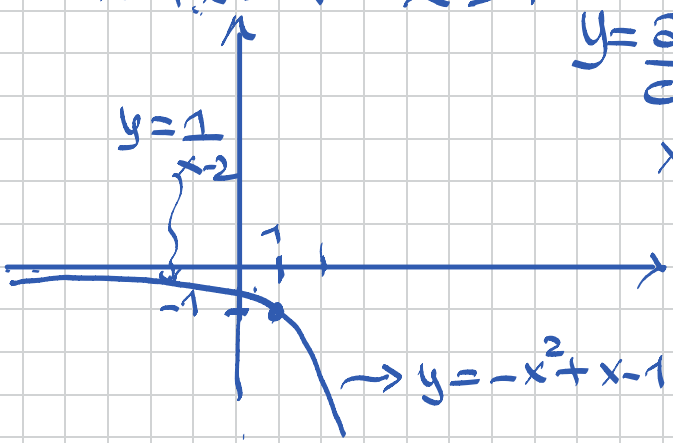
$$\begin{cases}a=2+b+1\\ b=a\end{cases} \qquad \begin{cases}b=-1\\ a=-1\end{cases}$$

$$f(x)=\begin{cases}\dfrac{-1}{2-x} & x<1\\ -1 & x=1\\ -x^2+x-1 & x>1\end{cases}$$

$y=\dfrac{1}{x-2}$

$V\left(-\dfrac{b}{2a};-\dfrac{\Delta}{4a}\right)$

$V\left(\dfrac{1}{2};-\dfrac{3}{4}\right)$



$y=\dfrac{ax+b}{cx+d}$

$x=-\dfrac{d}{c}$

$y=\dfrac{a}{c}$

$$f(x)=\begin{cases}\dfrac{\sin x^2}{x(\sqrt{x+1}-1)} & x>0\\ a\,2^x+3 & x\le 0\end{cases}$$

continuità nel suo dominio

$$f_1(x)=\frac{\sin x^2}{x(\sqrt{x+1}-1)}$$

$x\neq 0$

$x+1\geq 0 \quad x\geq -1$

$\sqrt{x+1}-1\neq 0$

$\sqrt{x+1}\neq 1 \quad x\neq 0$

definita $\forall x>0$

D: $\mathbb{R}$

continuità in $x=0$

$$\lim_{x\to 0^-} a\,2^x+3 = a+3$$

$$\lim_{x\to 0^+}\frac{\sin x^2}{x(\sqrt{x+1}-1)}=\lim_{x\to 0^+}\frac{x^2}{x\cdot\frac{1}{2}x}=2$$

$f(0)=a+3 \Rightarrow a+3=2$

$a=-1$

Stabilire quali tra le seguenti funzioni hanno una discontinuità eliminabile in $x=0$

1) $f(x) = \arctan\left(\frac{1}{x^3}\right)$

2) $g(x) = \frac{1}{x\sqrt{1+\frac{1}{x^2}}}$

3) $h(x) = \log|\sin x^2| - \log x^2$

1) $\lim_{x\to 0^-} \arctan\left(\frac{1}{x^3}\right) = -\frac{\pi}{2}$

$\lim_{x\to 0^+} \arctan\left(\frac{1}{x^3}\right) = \frac{\pi}{2}$

2) $\lim_{x\to 0^+} \frac{1}{x\sqrt{1+\frac{1}{x^2}}} = \lim_{x\to 0^+} \frac{1}{x\sqrt{\frac{1}{x^2}}}$

$= \lim_{x\to 0^+} \frac{1}{x\cdot\frac{1}{x}} = 1$

$\lim_{x\to 0^-} \frac{1}{x\sqrt{1+\frac{1}{x^2}}} = \lim_{x\to 0^-} \frac{1}{x\sqrt{\frac{1}{x^2}}}$

$\lim_{x\to 0^-} \frac{1}{x\left(-\frac{1}{x}\right)} = -1$ $\quad \left|\frac{1}{x}\right| = -\frac{1}{x}$

3) $\lim_{x\to 0^{\pm}} \log|\sin x^2| - \log x^2$

$\lim_{x\to 0^{\pm}} \log \frac{|\sin x^2|}{x^2} = \lim_{x\to 0} \log \frac{|x^2|}{x^2} =$

$= 0$ disc. eliminabile

$$f(x) = \begin{cases} \log|\sin x^2| - \log x^2 & x \neq 0 \\ 0 & x = 0 \end{cases}$$

Determinare l'insieme di definizione e classificare i punti di discontinuità di

$$f(x) = \begin{cases} 2^{\frac{1}{x-1}} - 2^{\frac{1}{x^2(x-1)}} & x \neq 0, 1 \\ 0 & x = 0, \ x = 1 \end{cases}$$

D: $\mathbb{R}$

$\lim_{x\to 0^{\pm}} 2^{\frac{1}{x-1}} - 2^{\frac{1}{x^2(x-1)}} = \frac{1}{2}$

DISC. ELIMINABILE



$\lim_{x\to 1^-} 2^{\frac{1}{x-1}} - 2^{\frac{1}{x^2(x-1)}} = 0$

$\frac{1}{x-1} \to -\infty$, $\frac{1}{x^2(x-1)} \to -\infty$

$2^{\frac{1}{x-1}} \to 0$, $2^{\frac{1}{x^2(x-1)}} \to 0$

$$\lim_{x\to 1^+} \; 2^{\frac{1}{x-1}} - 2^{\frac{1}{x^2(x-1)}}$$

$$\lim_{x\to 1^+} \; 2^{\frac{1}{x-1}} \left(1 - 2^{\frac{1}{x^2(x-1)} - \frac{1}{x-1}}\right)$$

$$\lim_{x\to 1^+} \; 2^{\frac{1}{x-1}} \left(1 - 2^{\frac{1-x^2}{x^2(x-1)}}\right)$$

$$\lim_{x\to 1^+} \; 2^{\frac{1}{x-1}} \left(1 - 2^{\frac{-(x+1)(x-1)}{x^2(x-1)}}\right) = +\infty$$

$x=1$ Disc. II^ specie

Data $f(x) = \begin{cases} e^{-x} & x<0 \\ 2+a(x-3) & 0 \le x < 3 \\ \frac{1}{(x-b)^3} - 1 & x \ge 3 \end{cases}$

Determinare $a, b \in \mathbb{R}$ in modo che sia applicabile il Teorema di Weierstrass nell'intervallo $[-1,3]$. Calcolare poi il massimo e il minimo assoluti in questo intervallo

$[-1,3]$ chiuso

Continuità in $x=0$ $\quad f(0) = 2-3a$

$\lim_{x\to 0^-} e^{-x} = 1 \qquad \lim_{x\to 0^+} 2+a(x-3) = 2-3a$

$2-3a=1 \quad \boxed{a=\frac{1}{3}}$

Continuità in $x=3$

$\lim_{x\to 3^-} 2+\frac{1}{3}(x-3)=2$

$\lim_{x\to 3^+} \frac{1}{(x-b)^3}-1 = \frac{1}{(3-b)^3}-1 = f(3)$

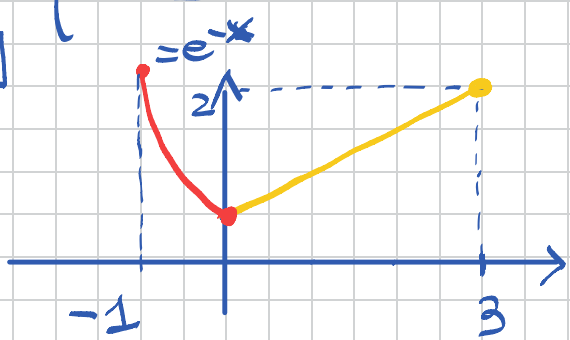
$\frac{1}{(3-b)^3}-1=2 \qquad \frac{1}{(3-b)^3}=3$

$(3-b)^3=\frac{1}{3} \qquad 3-b=\sqrt[3]{\frac{1}{3}}$

$\Rightarrow \boxed{b=3-\sqrt[3]{\frac{1}{3}}}$

$$f(x)=\begin{cases} e^{-x} & x<0 \\ 2+\frac{1}{3}(x-3) & 0\le x\le 3 \end{cases}$$

$[-1,3]$



$y=\frac{1}{3}x+1$

Min ass. $x=0 \quad f(0)=1$

max assoluto $x=-1 \quad f(-1)=e$

1) Stabilire se l'equazione

$$xe^x - \sin x = 2x^2 e^x (x-\pi) + 1$$

ammette soluzioni in $[0,\pi]$

$g(x) = xe^x - \sin x - 2x^2 e^x (x-\pi) - 1$

$g(x) = 0$

$g(x)$ continua in $[0,\pi]$

$g(0) \cdot g(\pi) < 0$

$g(0) = -1 < 0 \quad g(\pi) = \pi e^\pi - 1 > 0$

$\Rightarrow \exists$ uno zero di $g(x)$ in $[0,\pi]$

$\Rightarrow$ l'equazione ammette una soluzione

Mostrare che le funzioni

$f(x) = e^x + (x-1)\sin x$

$g(x) = 4 \arctan \frac{2x}{1+x^2}$

assumono lo stesso valore in almeno un punto di $[0,1]$

$h(x) = f(x) - g(x)$

$h(x) = e^x + (x-1)\sin x - 4 \arctan \frac{2x}{1+x^2}$

$h(x)$ continua in $[0,1]$

$h(0) = 1$

$h(1) = e - 4\arctan 1 = e - 4 \cdot \frac{\pi}{4} = e - \pi < 0$

$\Rightarrow \exists$ uno zero per $h(x)$ in $[0,1]$

$\Rightarrow f(x) = g(x)$ in almeno un punto di $[0,1]$

Dire se sono vere o false le affermazioni

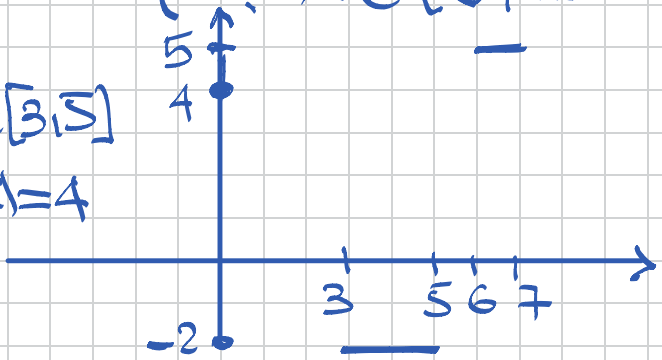
1) Sia $f: [3,5] \cup [6,7] \to \mathbb{R}$ continua tale che $f(3) = -2$, $f(7) = 5$

Allora a) $\exists\, x \in [3,5]$ tale che $f(x) = 4$ FALSO

b) non possiamo dire se la funzione si annulla VERO

CONTROESEMPIO per l'affermaz.

a) $f(x) = \begin{cases} -2 & x \in [3,5] \\ 5 & x \in [6,7] \end{cases}$



$\nexists\, x \in [3,5]$ t.c. $f(x) = 4$